

State of the Web 2020-2025: HTTP Archive Trends in Page Weight, Protocols, and Security

● For web performance engineers & front-end architects

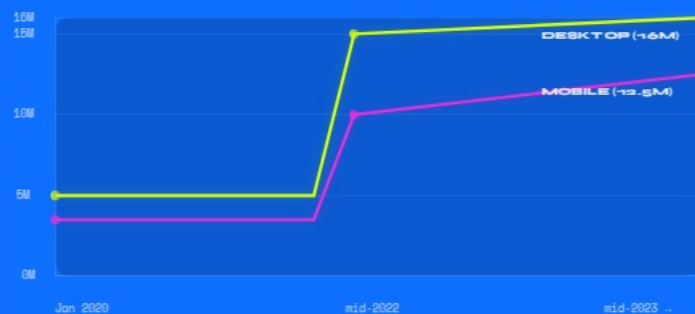
● Macro trends for benchmarking

Crawl Sample Size Tripled: From ~5M to ~16M Desktop URLs Since 2020

Desktop grew from roughly 5M URLs (Jan 2020) to ~16M by mid-2023 onward. Mobile tracks below: ~3.5M → ~12.5M.

Dual-line time series (schematic)

Desktop vs Mobile



Desktop URLs Mobile URLs

Values shown are the stated baselines and plateaus from the HTTP Archive "State of the Web" report summary.

What changed

Interpretation

- Scale matters:** sampling expanded to ~16M desktop / ~12.5M mobile URLs by mid-2023 onward.
- Mid-2022 discontinuity:** a sharp jump (~5M→~16M desktop; ~3.5M→~12.5M mobile) is visible in the report.
- Benchmarking note:** trends should be compared with awareness of the sample-size step change.

Median Page Weight Climbs Steadily to ~2,000 KB on Desktop

Median transfer size rose from ~1,600 KB (early 2020) to ~2,000 KB (now). Mobile tracks slightly lower (~1,900 KB). The 75th-percentile desktop page exceeds 4,000 KB.

Reference figure

Total KB time series



Figure: Total Kilobytes (desktop vs mobile) with IQR bands (as provided in the brief).

Key points

From brief

~2,000 KB

Desktop median (up from ~1,600 KB in early 2020; ▲ ~25%).

~1,900 KB

Mobile median (tracks slightly lower; ▲ ~25%).

> 4,000 KB

75th-percentile desktop page size (long tail of heavy pages).

76 Requests per Desktop Page While TCP Connections Drop to 12

Median requests: 76 (desktop) / 71 (mobile), each up ~3%. Meanwhile, median TCP connections fell from ~15 to 12 (desktop) and to 11 (mobile), reflecting multiplexing benefits.

Request counts (median)

Latest snapshot

76 req/page

Median resources requested on desktop pages.

71

Mobile median requests per page.

+3%

Increase over the observation period (desktop ▲ 2.7%, mobile ▲ 2.9%).

All values from the brief's "Page Weight & Requests" table.

TCP connections per page (median)

Declining trend



Brief notes: connections decline despite rising request counts—consistent with HTTP/2/HTTP/3 multiplexing.

HTTPS Requests Reach 99%: Encryption Is Now the Default

HTTPS is effectively universal: 99.2% of desktop requests and 99.1% of mobile requests use HTTPS—up roughly 19–20 percentage points since early 2020 (around 80%).

99%+

Encrypted by default – plaintext HTTP is effectively gone for requests in the dataset.

99.2%

Desktop requests over HTTPS.

99.1%

Mobile requests over HTTPS.

From brief: ~80% in early 2020 → 99%+ now (▲ ~19–20 pts).

Small time series (schematic)

~80% → 99%+



Endpoints from the brief; intermediate values are not enumerated.

HTTP/2 Holds at 58% While HTTP/3 Support Surges to 40%

Snapshot: HTTP/2 accounts for 58.6% (desktop) / 57.6% (mobile) of requests; HTTP/3 support is 38.7% (desktop) / 40.0% (mobile). HTTP Archive measures support (Alt-Svc) rather than actual usage (fresh Chrome defaults to HTTP/2 first).

Latest snapshot

Grouped bars



Values from the brief's protocol table.

HTTP/3 rise (schematic)

Near-zero → ~40%



Measurement caveat: HTTP Archive reports HTTP/3 *support*, not necessarily negotiated usage, due to fresh Chrome connection behavior.

Font-Display Usage Reaches 42% on Mobile, Up 61% Since 2020

``font-display`` helps avoid Flash of Invisible Text (FOIT). Adoption (Lighthouse): 40.1% desktop and 42.5% mobile. Mobile is up 61.0% since 2020 (desktop up 14.9%).

Latest snapshot

Desktop vs Mobile



Measured by Lighthouse; values from the brief.

Why it matters

FOIT avoidance

- **Adoption:** 40.1% desktop / 42.5% mobile pages use ``font-display``.
- **Growth:** up 14.9% (desktop) and 61.0% (mobile) since 2020.
- **Opportunity:** more than half of pages still risk FOIT.

Benchmark

If you rely on web fonts, compare your site's behavior against a ~40-42% ecosystem baseline.

Key Takeaways: Bigger Pages, Smarter Protocols, Stronger Security

Five baselines to carry into your own performance reviews and infrastructure decisions.

- **Page weight keeps climbing:** desktop median ~2,000 KB (up from ~1,600 KB; ▲ ~25%); mobile ~1,900 KB.
- **HTTPS is effectively universal:** 99.2% desktop / 99.1% mobile requests over HTTPS (from ~80% in early 2020).
- **HTTP/3 support is the fastest mover:** ~38.7% desktop / 40.0% mobile support from near-zero, with the “support vs usage” caveat.
- **Connections are dropping:** median TCP connections per page fell from ~15 to 12 (desktop) / 11 (mobile), despite 76/71 requests per page.
- **Font-display still underused:** 40.1% desktop / 42.5% mobile adoption—more than half of pages still risk FOIT.

Call to action

Benchmark

Use these as ecosystem baselines: compare your median transfer size, request counts, connection counts, protocol mix, and font loading behavior—and prioritize the deltas with the largest user-impact.